

Mastering Embedded Systems Online Diploma

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FIRST TERM (FINAL PROJECT 2)

ENG. MAHMOUD ELYAMANI

MY PROFILE:

www.learn-in-depth-store.com/profile/mahmoud-elyamani-7/profile

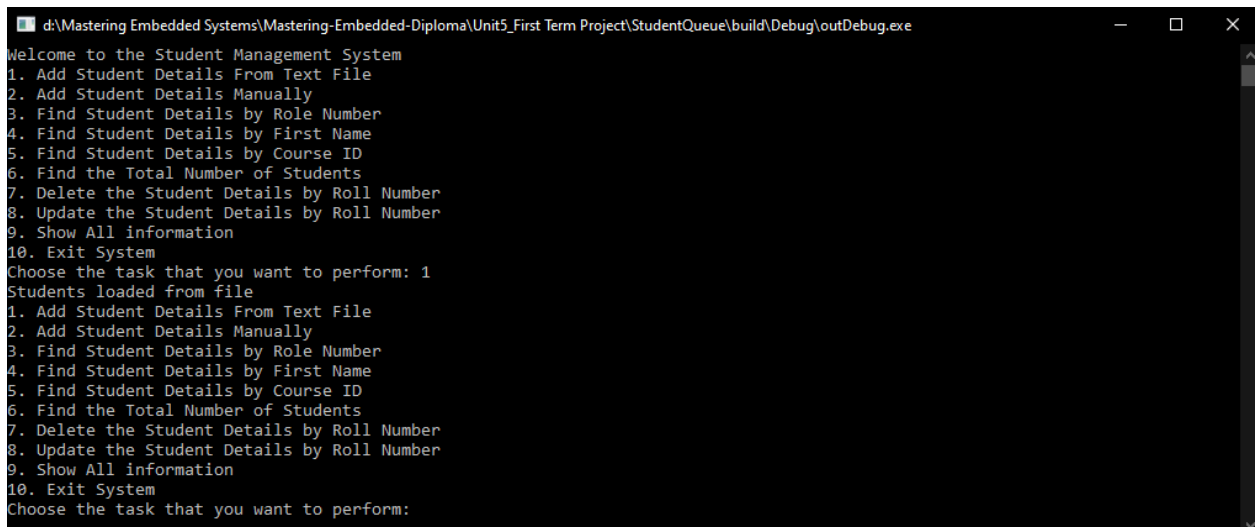
.Txt file:



```
students - Notepad
File Edit Format View Help
1 Mahmoud Elyamani 3.4 100 200 300 400 500
2 Keroles Karam 3.6 210 111 621 133 111
3 Ahmed Ali 3.5 101 102 103 104 105
4 Sara Mohamed 3.8 201 202 203 204 205
```

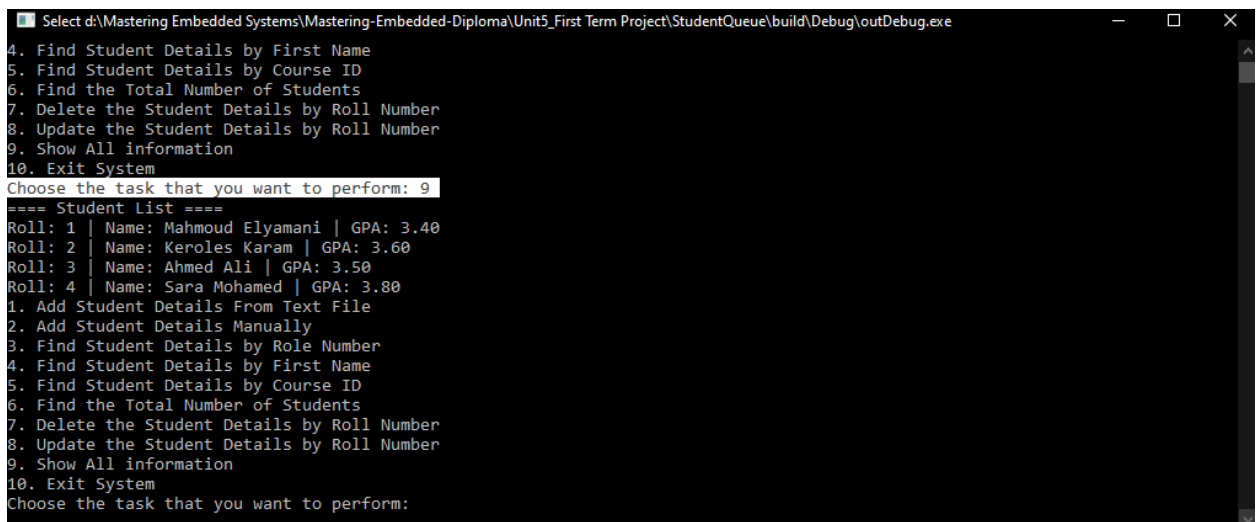
- For code preview please check out [Github Link to my project](#) as it is too big to paste screenshots of it here.

Case(1): Adding students details from .txt file:



```
d:\Mastering Embedded Systems\Mastering-Embedded-Diploma\Unit5_First Term Project\StudentQueue\build\Debug\outDebug.exe
Welcome to the Student Management System
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform: 1
Students loaded from file
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

Viewing the students after adding:



```
Select d:\Mastering Embedded Systems\Mastering-Embedded-Diploma\Unit5_First Term Project\StudentQueue\build\Debug\outDebug.exe
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform: 9
==== Student List ====
Roll: 1 | Name: Mahmoud Elyamani | GPA: 3.40
Roll: 2 | Name: Keroles Karam | GPA: 3.60
Roll: 3 | Name: Ahmed Ali | GPA: 3.50
Roll: 4 | Name: Sara Mohamed | GPA: 3.80
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

Case(2): Adding a student manually through the terminal

```
Select d:\Mastering Embedded Systems\Mastering-Embedded-Diploma\Unit5_First Term Project\StudentQueue\build\Debug\outDebug.exe
10. Exit System
Choose the task that you want to perform: 2
Enter Roll Number: 8
Enter First Name: Marco
Enter Last Name: Magdy
Enter GPA: 3.3
Enter 5 Course IDs: 55
66
77
88
99
Student added successfully
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

Viewing changes:

```
Select d:\Mastering Embedded Systems\Mastering-Embedded-Diploma\Unit5_First Term Project\StudentQueue\build\Debug\outDebug.exe
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform: 9
==== Student List ====
Roll: 1 | Name: Mahmoud Elyamani | GPA: 3.40
Roll: 2 | Name: Keroles Karam | GPA: 3.60
Roll: 3 | Name: Ahmed Ali | GPA: 3.50
Roll: 4 | Name: Sara Mohamed | GPA: 3.80
Roll: 8 | Name: Marco Magdy | GPA: 3.30
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

Case(3): Reviewing student's details by role number:

```
Select d:\Mastering Embedded Systems\Mastering-Embedded-Diploma\Unit5_First Term Project\StudentQueue\build\Debug\outDebug.exe
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform: 3
Enter Roll Number: 2
Found: Kerolos Karam | GPA: 3.60
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

Case(4): Reviewing student's details by first name:

```
Select d:\Mastering Embedded Systems\Mastering-Embedded-Diploma\Unit5_First Term Project\StudentQueue\build\Debug\outDebug.exe
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform: 4
Enter First Name: Ahmed
Roll: 3 | GPA: 3.50
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

Case(5): Reviewing student's details by course ID:

```
10. Exit System
Choose the task that you want to perform: 5
Enter Course ID: 55
Roll: 8 | Name: Marco
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

Case(6): Finding the total number of students

```
Select d:\Mastering Embedded Systems\Mastering-Embedded-Diploma\Unit5_First Term Project\StudentQueue\build\Debug\outDebug.exe
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform: 6
Total Students = 5
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

Case(7): Deleting a student then checking the total again

```
Choose the task that you want to perform: 7
Enter Roll Number to delete: 4
Student deleted
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

```
Choose the task that you want to perform: 6
Total Students = 4
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

```
Choose the task that you want to perform: 9
==== Student List ====
Roll: 1 | Name: Mahmoud Elyamani | GPA: 3.40
Roll: 2 | Name: Keroles Karam | GPA: 3.60
Roll: 3 | Name: Ahmed Ali | GPA: 3.50
Roll: 8 | Name: Marco Magdy | GPA: 3.30
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

Case(8): Updating student's info by role number

```
Select d:\Mastering Embedded Systems\Mastering-Embedded-Diploma\Unit5_First Term Project\StudentQueue\build\Debug\outDebug.exe
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform: 8
Enter Roll Number to update: 1
Enter new First Name: Elyamani
Enter new Last Name: Mahmoud
Enter new GPA: 4
Updated successfully
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

Case(9): Showing all students after editing

```
Select d:\Mastering Embedded Systems\Mastering-Embedded-Diploma\Unit5_First Term Project\StudentQueue\build\Debug\outDebug.exe
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform: 9
==== Student List ====
Roll: 1 | Name: Elyamani Mahmoud | GPA: 4.00
Roll: 2 | Name: Keroles Karam | GPA: 3.60
Roll: 3 | Name: Ahmed Ali | GPA: 3.50
Roll: 8 | Name: Marco Magdy | GPA: 3.30
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```

Case(10): attempting to view a non-existent role number:

```
Choose the task that you want to perform: 3
Enter Roll Number: 99
Error!Student not found
1. Add Student Details From Text File
2. Add Student Details Manually
3. Find Student Details by Role Number
4. Find Student Details by First Name
5. Find Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show All information
10. Exit System
Choose the task that you want to perform:
```